What is a force, really?

? The first law is recited as "a body keeps moving unless pushed". Why is it a definition of the frame rather than a law about forces?

1 A frame you are allowed to use

An inertial frame is one in which a body with no net force keeps a constant velocity. That is a test you can run with a puck on a level air table, and it is also a definition: the first law selects the frames in which F = ma may be written.

So the law is not "motion needs a push". It is "no push means no change", and only in the right room.

2 The same ball in two books

In the ground frame the ball on a table has zero net force and zero acceleration. In the frame of a braking car the same ball drifts forward although nothing touched it: the frame invented a force.

Add the pseudo force $-ma_0$ and the bookkeeping is repaired, which is what the last section of this part does properly.

→ a frame that needs a fix-up force to explain free motion is not inertial

3 How to detect one without looking out

A plumb line hangs off vertical, a water surface tilts, a pendulum swings with a changed period. No sealed box can distinguish "I am accelerating" from "a uniform gravitational field appeared" — that is the equivalence principle speaking.

$$a_{\text{frame}} \neq 0 \;\Rightarrow\; \text{pseudoforcesappearineveryequation}$$

4 A worked test with real numbers

A puck slides on a level air table at 3 m/s while the room is at rest. A second puck is released inside a van braking at 2 m/s². What does each observer measure?

$$\text{ground}: \; a = 0 \quad \text{van}: \; a = 2 \text{ m/s}^2 \text{ forward}$$

→ the van observer must invent f = ma = 2m forward; the ground observer invents nothing

DEFINE

inertial frame

A frame in which a force-free body has constant velocity, and in which F = ma holds without correction terms.

TRAP

the first law is not F = 0 at rest

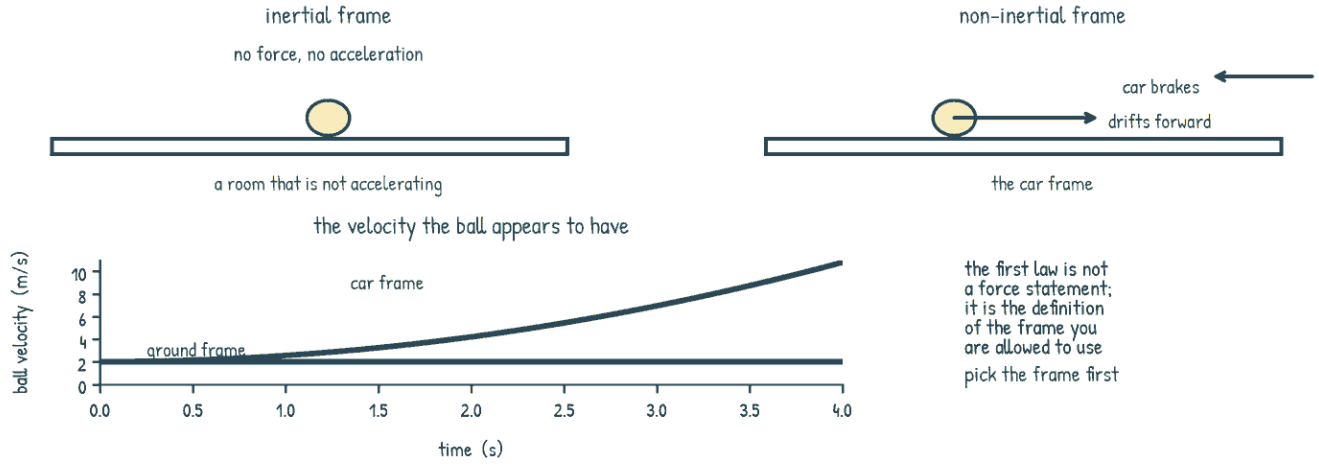
A body moving uniformly has zero net force. Rest is not privileged; constant velocity is.

CHECK

one-line test

If you had to invent a force to explain a motion, you are in a non-inertial frame or you forgot an interaction.

Pick the frame first: in an inertial frame a force-free body keeps its velocity



the first law is not a force statement; it is the definition of the frame you are allowed to use

pick the frame first

KEEP / pick the frame first: a force is what an inertial frame says it is

Is F = ma the law, or a special case?

? Rockets, falling chains and conveyor belts all have mass entering or leaving. Which form of Newton's second law survives all three?

1 The statement that always holds

Force is the rate of change of momentum. For a fixed mass this collapses to ma, which is why the textbook form works almost everywhere — and why it fails exactly where mass is changing.

$$\vec{F} = \frac{d\vec{p}}{dt} = m\vec{a} + \vec{v}\frac{dm}{dt}$$

2 A hit is an area, not a number

In a collision the force is a pulse lasting a millisecond or less. What survives is its integral: the impulse, equal to the momentum change. The area under an F–t graph is the whole story.

$$\vec{J} = \int_{t_1}^{t_2} \vec{F}\,dt = \Delta\vec{p}$$

→ if the pulse is 1000 N for 1 ms, J = 1 N s whether or not you know its shape

3 When mass arrives already moving

Sand dropped on a belt has zero horizontal momentum; the belt must give it v. The force is $\dot{m}v$, not ma, because dm/dt is doing the work.

$$F = \frac{dp}{dt} = \dot{m}\,v \qquad \text{for mass arriving from rest}$$

→ the same term appears as λv^2 when a chain is picked up off the floor

4 A worked impulse budget

A 0.145 kg cricket ball arrives at 30 m/s and leaves at 40 m/s along the same line, contact lasting 1.2 ms. Find the impulse and the mean force.

$$J = m\Delta v = 0.145(40 + 30) = 10.2 \text{ N s}$$

$$\bar{F} = \frac{J}{\Delta t} = \frac{10.2}{1.2 \times 10^{-3}} = 8.5 \text{ kN}$$

→ the ball's weight, 1.4 N, is 0.017 % of that: gravity can be dropped during the hit

KEY RESULT

three forms

$$\vec{F} = m\vec{a}$$

$$\vec{F} = \frac{d\vec{p}}{dt}$$

$$\vec{J} = \Delta\vec{p}$$

Use the second whenever mass changes; the first is a corollary.

TRAP

which mass

In F = ma, m is the mass being accelerated right now. For a rocket it is m(t), and the exhaust momentum must be counted separately.

NUMBERS

feel the scale

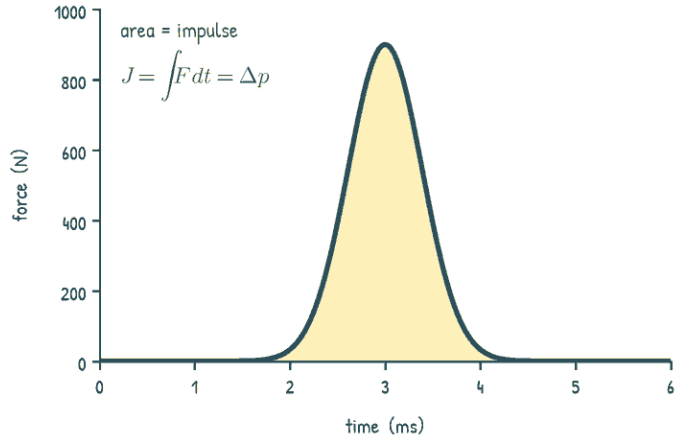
1000 N for 1 ms = 1 N·s

a 0.145 kg ball reversed at 7 m/s needs 2 N·s

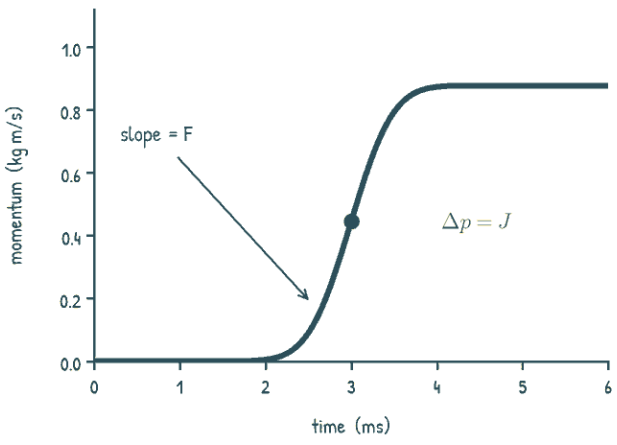
$\dot{m}$ = 2 kg/s at 3 m/s needs 6 N

F = dp/dt is the general law; F = ma is the fixed-mass shortcut, and the impulse is the area

a real hit is a pulse, not a number



the curve of momentum is what F shapes



KEEP / F = dp/dt is the law; F = ma is the fixed-mass shorthand

Why is the normal force not the reaction to weight?

? Equal and opposite sounds like a pair. What extra conditions turn two forces into a third-law pair?

1 Four conditions, not two

A third-law pair is equal in magnitude, opposite in direction, of the same type, and acts on two different bodies. It comes from one interaction, so it dies when the interaction dies.

Push hard on a wall and the wall pushes back just as hard — until you let go, when both forces vanish together.

2 The block on the table

The table pushes the block up with N. The block pushes the table down with an equal N. That is the pair. The block's weight pairs with the block pulling the Earth up, a force nobody draws.

$$N_{\text{onblock}} \;\leftrightarrow\; N_{\text{ontable}}$$

$$mg_{\text{onblock}} \;\leftrightarrow\; mg_{\text{onEarth}}$$

→ N and mg act on the same body and are different in type: never a pair

3 Why they cancel only sometimes

On a table at rest, N = mg and the two cancel, which hides their different origins. In a lift accelerating upward N > mg, and in free fall N = 0 while mg is unchanged.

$$N - mg = ma \;\Rightarrow\; N = m(g + a)$$

4 A worked pair audit

A 2 kg block rests on a table. List the forces on the block and on the table, then pair them.

$$\text{block}: \; N = 19.6\ \text{N}\uparrow,\; mg = 19.6\ \text{N}\downarrow$$

$$\text{table}: \; 19.6\ \text{N}\downarrow \text{ from block},\; 19.6\ \text{N}\uparrow \text{ from floor}$$

→ the third-law partner of N on the block is the 19.6 N push on the table, not mg

DEFINE

third-law pair

Same type, equal size, opposite direction, different bodies, one interaction.

TRAP

the popular wrong pair

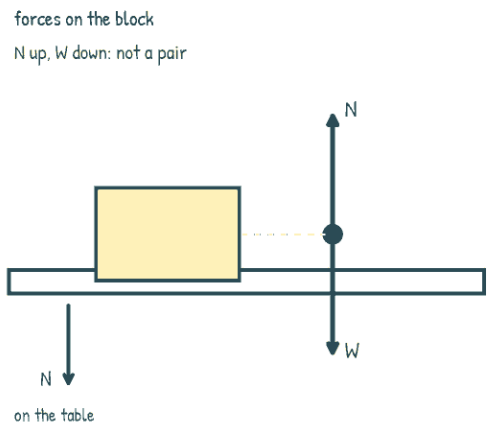
N and mg are not a pair: same body, different interaction. They balance only when a = 0.

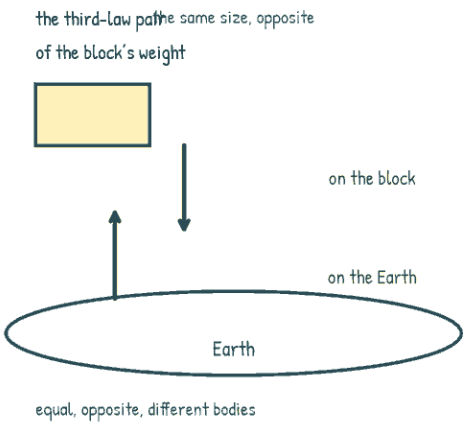
CHECK

name both bodies

Say the sentence out loud: "the table pushes the block, the block pushes the table." If the two names are not different, it is not a pair.

Equal and opposite, but never on the same body: the normal force is a contact pair, not a weight pair





KEEP / a pair needs two bodies and one interaction; N and mg are neither

How do you draw a diagram that never lies?

? Six forces, one acceleration. What procedure makes the equations fall out in one pass?

1 ## Isolate exactly one body

Draw the body alone. Every contact becomes a normal and possibly a friction; every rope becomes a tension along the rope; every mass gets its weight. Nothing else is allowed in.

If you cannot name the agent that exerts a force, erase the force. "Centrifugal" and "the force of motion" fail this test immediately.

2 ## Choose axes along the acceleration

On an incline the acceleration is along the slope, so resolve along and perpendicular to it. The two equations then decouple: one gives N, the other gives a.

$$mg\sin\theta - f = ma$$

$$N = mg\cos\theta$$

→ N is less than mg on a slope, equal to mg only on level ground

3 ## The two errors that survive to the exam

Counting the centripetal force as an extra arrow, and counting the pseudo force without declaring a non-inertial frame. Both add a force that is already the resultant.

→ also: a force on the body you did not isolate belongs somewhere else

4 ## A worked incline

A 5 kg block slides down a 30° rough incline with μ_k = 0.2. Find N, the friction, and the acceleration.

$$N = mg\cos 30^\circ = 42.4\ \text{N}, \qquad f = \mu_k N = 8.5\ \text{N}$$

$$a = g(\sin 30^\circ - \mu_k \cos 30^\circ) = 9.8(0.5 - 0.173) = 3.2\ \text{m/s}^2$$

→ N is 42.4 N, not the 49 N weight: that difference is the whole point of resolving

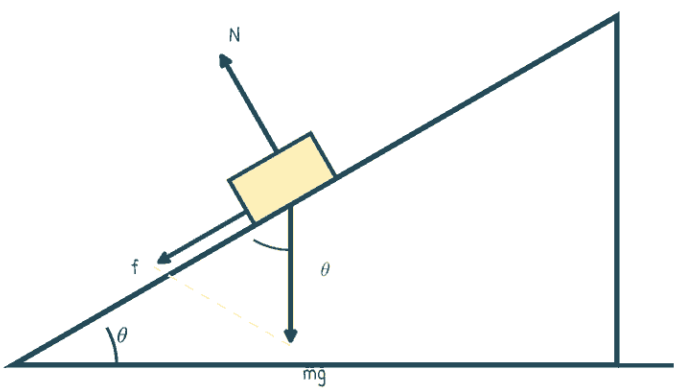
GATE
the five steps
1 isolate 2 contacts 3 ropes 4 weight 5 axes along a
Then write ΣF = ma twice, on the two axes.

TRAP
double counting
Centripetal force is the name of the inward resultant, not a fifth arrow. Add it and you have solved 2mv²/R = 2mv²/R.

CHECK
sanity pass
Zero acceleration must reproduce the statics answer. Set a = 0 and see.

Isolate the body, draw only external forces, put the axes along the acceleration

isolate, then rotate the axes



$mg\sin\theta - f = ma$

$N = mg\cos\theta$

axes along the acceleration

decouple the two equations

$N \neq mg$ unless $\theta = 0$

KEEP / isolate one body, name every agent, put the axes along the acceleration

What does a weighing scale actually measure?

? In a lift the scale reads more, less, or nothing at all. Which force is it reporting?

1 The four standard forces

Weight mg, always vertically down, always the same mg. Normal N, perpendicular to the surface, can only push. Tension T, along the string, can only pull. Spring force kx, toward the natural length.

None of these is defined by a formula that always holds: N is whatever the constraint requires, up to the point where contact is lost.

2 The scale reads N, not mg

A spring scale measures the force squeezing it, which is the normal force on your feet. Apply Newton to the person:

$$N - mg = ma \;\Rightarrow\; N = m(g + a)$$

$$\text{liftup} : m(g+a) \quad \text{freefall} : 0$$

→ accelerating down at a < g gives a reading of m(g − a), still positive

3 Weightless is not gravity-free

An astronaut in orbit still feels 8.7 m/s² of gravity. Everything around falls with the same acceleration, so nothing presses and N = 0. The scale cannot tell weightlessness from free fall, because they are the same thing.

4 A worked lift ride

A 60 kg passenger stands on a scale in a lift that accelerates up at 2 m/s², then brakes at 2 m/s². What does the scale read?

$$N = m(g+a) = 60(11.8) = 708 \text{ N}$$

$$N = m(g-a) = 60(7.8) = 468 \text{ N}$$

→ true weight is 588 N throughout; only the reading changes

KEY RESULT

reading on a scale

$$N = m(g+a) \;\; (\text{uppositive})$$

up at g/2 reads 1.5 mg; down at g/2 reads 0.5 mg

TRAP

"my weight changed"

mg is fixed by mass and the local field. Only N changes. A scale reads N.

NUMBERS

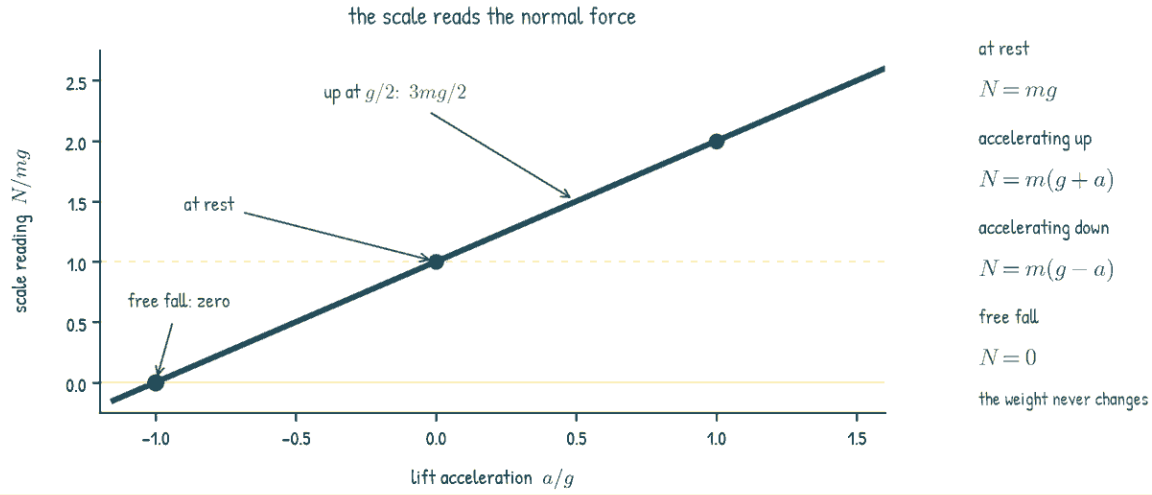
quick numbers

a = g: double weight, zero weight

a = g/3 up: 1.33 mg

spring balance in a lift reads the tension, not mg

Weight never changes; the reading is the normal force, and free fall makes it zero



at rest

$$N = mg$$

accelerating up

$$N = m(g+a)$$

accelerating down

$$N = m(g-a)$$

free fall

$$N = 0$$

the weight never changes

KEEP / weight never changes; the reading is the normal force

Three forces, no acceleration: why does the triangle close?

? A lamp hangs on two wires. When is the sine rule faster than resolving components?

1

Equilibrium is a vector statement

Zero net force means the three arrows, placed head to tail, return to the start. That closed triangle is the whole content of the equilibrium condition; components are just one way to read it.

$$\vec{F}_1 + \vec{F}_2 + \vec{F}_3 = 0$$

2

Lami is the sine rule in disguise

In the force triangle, side F_1 is opposite the angle between F_2 and F_3. The sine rule then gives all three ratios at once.

$$\frac{F_1}{\sin\alpha_1} = \frac{F_2}{\sin\alpha_2} = \frac{F_3}{\sin\alpha_3}$$

→ α_1 is the angle between F_2 and F_3 — the one opposite F_1

3

Where it stops working

Four or more forces: the polygon still closes, but there is no single sine-rule shortcut. A couple or a body with size: force equilibrium alone is not enough, and moments must be added.

4

A worked lamp

A 100 N lamp hangs from two wires making 30° and 60° with the ceiling. Find both tensions by Lami.

$$\frac{T_1}{\sin 120^\circ} = \frac{T_2}{\sin 150^\circ} = \frac{100}{\sin 90^\circ}$$

$$T_1 = 86.6\ \text{N}, \qquad T_2 = 50.0\ \text{N}$$

→ resolving gives the same pair, in twice the lines

KEY RESULT

Lami

$$\frac{F_1}{\sin\alpha_1} = \frac{F_2}{\sin\alpha_2} = \frac{F_3}{\sin\alpha_3}$$

each force over the sine of the angle opposite it

TRAP

which angle

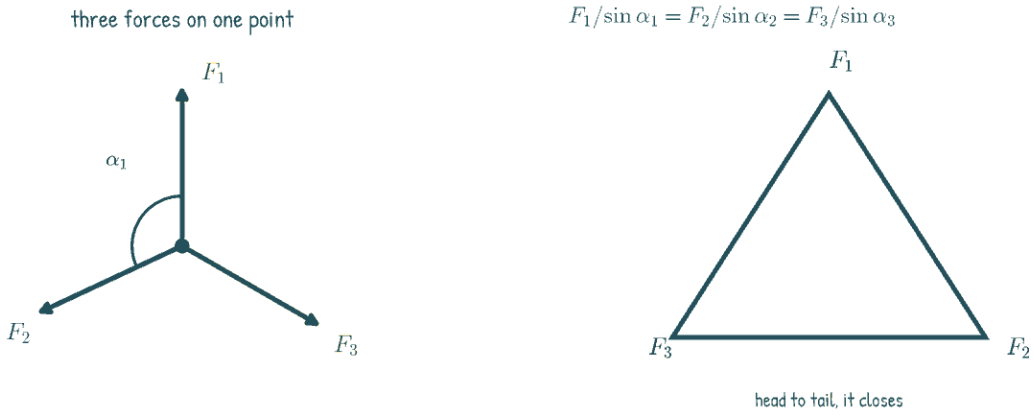
The angle is between the other two forces, not between the force and the horizontal. Using the complement gives a wrong ratio that looks plausible.

GATE

when to reach for it

Exactly three forces, all concurrent, body at rest or moving uniformly. Otherwise resolve.

Three forces, one point, zero sum: the vector triangle obeys the sine rule



KEEP / three forces in equilibrium close a triangle, and the sine rule reads it

Is friction equal to mu N?

? A 20 N block sits on a rough floor while you push with 3 N and it does not move. What is the friction force?

1

Static friction is a range

A block that has not started to slide has zero acceleration, so friction must exactly cancel the push. It takes whatever value is needed, from zero up to a ceiling set by the surfaces.

$$0 \le f_s \le \mu_s N$$

→ push with 3 N and it does not move: f_s = 3 N, not μN

2

At the ceiling it lets go

Increase the push and f rises with it, one for one, until f reaches μ_s N. That instant is the threshold of slipping, and the angle at which a ramp reaches it is the angle of repose.

$$\mu_s = \tan\theta_{\text{repose}}$$

3

Kinetic friction is a value

Once sliding, the force drops to a fixed μ_k N, usually a little below the static maximum. That drop is why a pushed wardrobe suddenly lurches, and why a car's ABS tries to stay just under the threshold.

$$f_k = \mu_k N \qquad \mu_k < \mu_s \text{ usually}$$

4

A worked push

A 4 kg crate sits on a floor with μ_s = 0.5. You push with 15 N, then with 25 N. What is the friction each time?

$$\mu_s N = 0.5(39.2) = 19.6 \text{ N}$$

$$F = 15 \text{ N}: \ f = 15 \text{ N} \qquad F = 25 \text{ N}: \text{ it slides}$$

→ at 25 N the friction drops to μ_k N, so the crate accelerates rather than creeping

KEY RESULT

the three statements

$$f_s \le \mu_s N, \quad f_s = \mu_s N \text{ at slip}$$

$$f_k = \mu_k N$$

$$\mu_s = \tan\theta_{\text{repose}}$$

TRAP

f = mu N on a stationary body

This is the single most common error in the chapter. On a body that is not slipping, friction is found from equilibrium, then checked against μ_s N.

NUMBERS

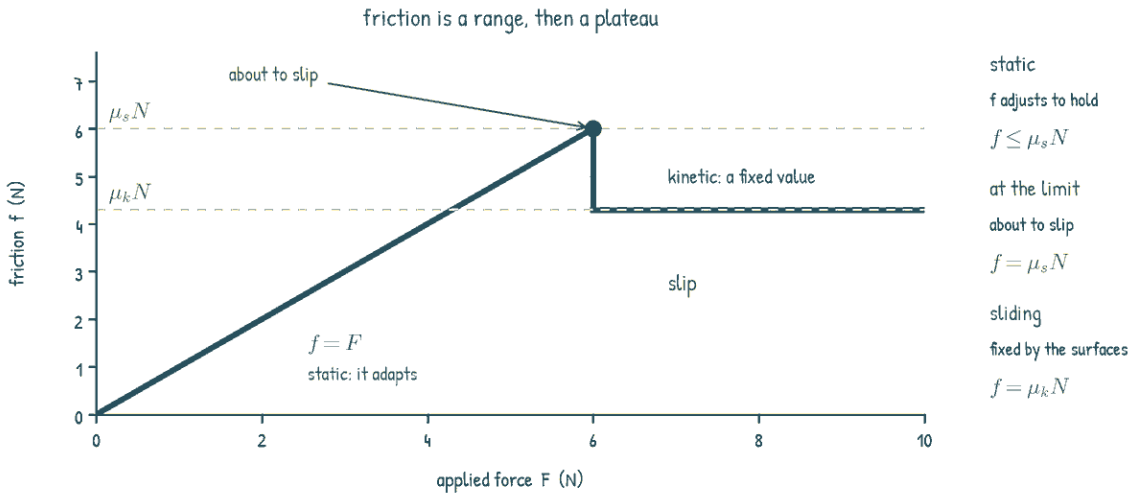
check these

μ_s = 0.6 on 2 kg: ceiling 11.8 N

push 6 N: f = 6 N, still static

push 13 N: it slides with f = μ_k N

Static friction is an inequality with a ceiling; kinetic friction is a value



static

f adjusts to hold

$$f \le \mu_s N$$

at the limit

about to slip

$$f = \mu_s N$$

sliding

fixed by the surfaces

$$f = \mu_k N$$

KEEP / static friction is an inequality with a ceiling, not a number

Which way does friction point?

? A block is dropped on a moving belt and another slides up a ramp. Both are moving, yet friction points in opposite senses. Why?

1 ## Compare the two surfaces, not the body

Friction acts at the interface. What matters is the velocity of the block relative to the surface it touches. If the block lags, friction drags it forward; if it leads, friction holds it back.

$$\vec{f} \text{ opposes } \vec{v}_{\text{block}} - \vec{v}_{\text{surface}}$$

2 ## Three standard pictures

Pushed on a floor: the block would slide right, friction points left. Dropped on a belt: the belt runs forward under it, friction points forward and the block speeds up. Sliding up a ramp: friction points down the slope.

→ on the belt the block's speed increases, so friction does positive work on it

3 ## Where this bites

Walk forward: the floor pushes you forward by friction. Accelerate a car: the road pushes the driving wheels forward. Brake: the same friction points backward on the car. In each case compare the tyre contact patch with the road.

4 ## A worked belt

A 2 kg package is dropped on a belt running at 4 m/s, with μ_k = 0.3. How long until it stops slipping, and how far does it travel?

$$a = \mu_k g = 2.94 \text{ m/s}^2, \qquad t = \frac{4}{2.94} = 1.36 \text{ s}$$

$$s_{\text{pkg}} = \tfrac{1}{2}(4)(1.36) = 2.72 \text{ m}, \qquad s_{\text{belt}} = 5.44 \text{ m}$$

→ the belt slides 2.72 m under the package, and that gap is the friction heat

GATE
the one question
What is the velocity of this body relative to the surface it touches? Friction points opposite to that.

TRAP
opposing the motion in the room
On an accelerating belt, on a turning car, on a walker: friction often points along the motion. It always opposes slip.

CHECK
sign of work
Friction can do positive work on one body and equal negative work on the other. The pair together always removes mechanical energy into heat.

Friction is about relative slip at the contact, not about the body's motion in the room

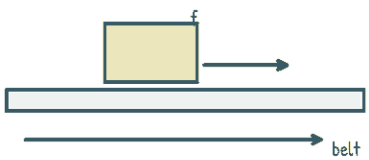
pushed on the floor

it would slide right,
so f points left



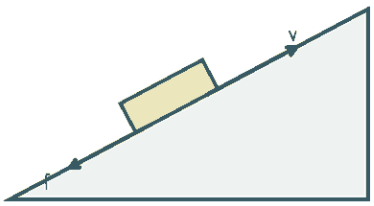
on a moving belt

the block lags behind,
so f points forward



sliding up a ramp

friction acts down
the slope



KEEP / friction opposes relative slip at the contact, not motion in the room

Why is the easiest pull not horizontal?

? Dragging a crate is easier with the rope tilted up. Why, and exactly how much less force is needed?

1

The trade

Pulling at angle θ upward reduces the normal force to mg – F sinθ, so it reduces the friction ceiling. But only the horizontal part F cosθ drives the crate. The two effects fight, so there is an optimum.

$$N = mg - F\sin\theta, \qquad f = \mu N$$

$$F\cos\theta = \mu(mg - F\sin\theta)$$

2

Minimise the force

Solve for F as a function of θ and minimise. The denominator cosθ + μ sinθ peaks at tanθ = μ, which fixes both the angle and the smallest force.

$$F(\theta) = \frac{\mu mg}{\cos\theta + \mu\sin\theta}$$

$$\theta^* = \tan^{-1}\mu, \qquad F_{\min} = \frac{\mu mg}{\sqrt{1+\mu^2}}$$

→ μ = 0.5 on 2 kg: 26.6° and 8.77 N instead of 9.8 N

3

The other use of the same algebra

Let θ grow to 90° and the crate lifts instead of sliding. The angle of repose is the mirror idea: on a ramp, tanθ ≤ μ_s keeps a block still without any external pull.

→ both results come from setting the friction ceiling equal to the driving component

4

A worked optimum

A 20 kg crate on μ = 0.4 needs to be dragged. Compare a horizontal pull with the optimum angle.

$$F_0 = \mu mg = 78.4\ \text{N}$$

$$F_{\min} = \frac{78.4}{\sqrt{1+0.16}} = 72.8\ \text{N at } 21.8^\circ$$

→ 5.6 N saved, and the crate unloads 27 N of normal force while being pulled

KEY RESULT

minimum pull

$$\tan\theta = \mu$$

$$F_{\min} = \mu mg/\sqrt{1+\mu^2}$$

the saving is only about 11 % at μ = 0.5, but it is free

NUMBERS

worked

m = 2 kg, μ = 0.5, g = 9.8

horizontal: F = 9.8 N

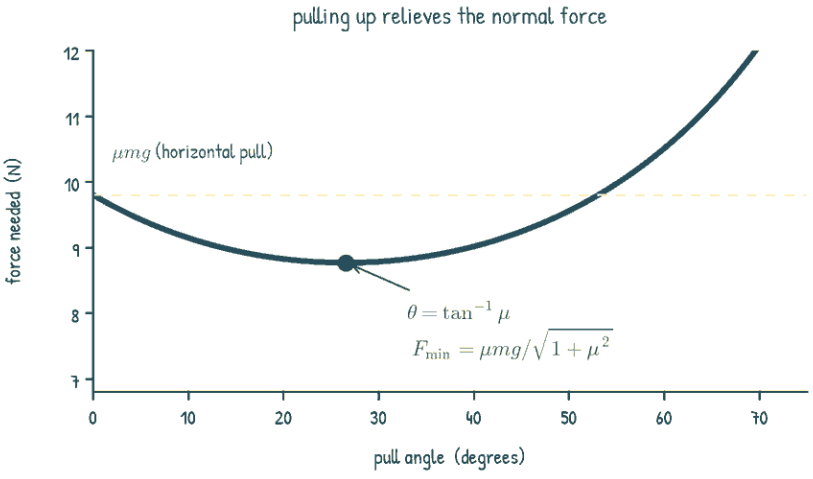
at 26.6°: F = 8.77 N

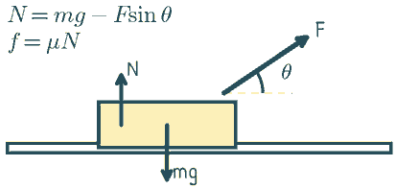
TRAP

two sign errors

N = mg – F sinθ (not plus) when you pull upward. And the friction term uses this reduced N, not mg.

Tilting the pull up steals normal force from the floor, and the best tilt is exactly arctan(mu)





KEEP / tilting up trades normal force for drive, and the best tilt is arctan mu

When do two stacked blocks stop moving together?

? You push the lower block of a stack. What limits how hard you can push before the top one slips?

1 **Look at the top block first**

The only horizontal force on the upper block is friction from the lower one. Its acceleration is therefore f/m, and it can never exceed μ_s g, no matter how hard you push below.

$$a_{\text{top}} = \frac{f}{m} \le \mu_s g$$

2 **While they move together**

Treat the pair as one body: a = F/(M + m). That is consistent with the top block only while the friction needed stays under the ceiling.

$$F \le \mu_s g(M+m)$$

$$\mu_s = 0.3,\ m = 1\ \text{kg},\ M = 2\ \text{kg} \Rightarrow F_{\max} = 8.82\ \text{N}$$

3 **After the break**

The top block keeps accelerating at μ_s g while the bottom one accelerates faster. The relative acceleration is constant, so the top block slides backward relative to the bottom — it is not "left behind" by inertia, it is dragged at a fixed maximum rate.

4 **A worked stack**

m = 2 kg on M = 3 kg, μ_s = 0.4 between them, floor frictionless. How hard can you push the lower block?

$$F_{\max} = \mu_s g(M+m) = 0.4(9.8)(5) = 19.6\ \text{N}$$

$$a_{\max} = \mu_s g = 3.92\ \text{m/s}^2$$

→ push 19.6 N and the pair moves at 3.92 m/s²; push 25 N and the top one lags

KEY RESULT

the two numbers

$$F_{\max} = \mu_s g(M+m)$$

$$a_{\max} = \mu_s g$$

the ceiling is set by the upper block, never by the push

TRAP

pushing vs pulling

Which block you push changes the answer: the limit above is for a push on the lower block. Repaint the free-body diagram for the other case.

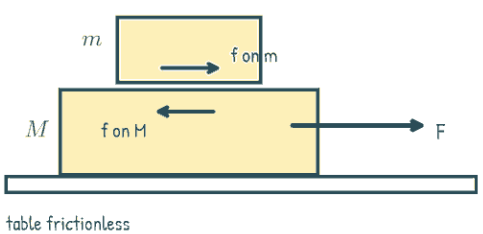
CHECK

limit case

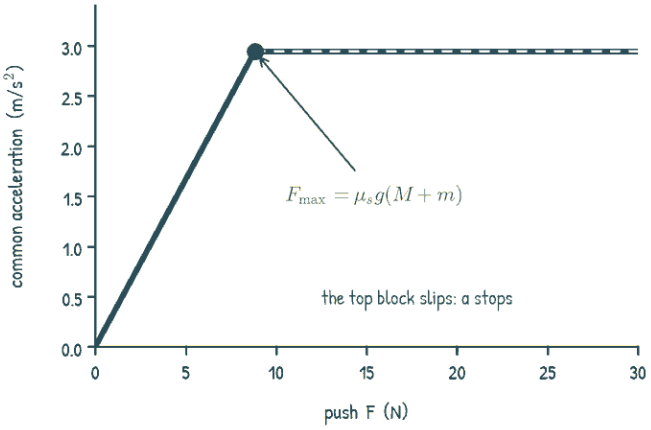
Set μ_s = 0 and the pair cannot move together at all: the top block stays put while the bottom slides out.

The top block is driven by friction alone, so it caps the whole system at a = mu_s g

the friction that drags the top block



together only up to a point



KEEP / the top block is driven by friction alone, so it caps the whole system

System or single body: which do you solve?

? Two blocks are pushed together. How do you get both the common acceleration and the force between them, without solving twice?

1

The system first

Internal forces come in third-law pairs, so they cancel inside a system. A single equation then gives the acceleration of the whole thing.

$$a = \frac{F}{m_1 + m_2}$$

2

Then isolate the far body

Pick the body that has the fewest forces — usually the one you are not pushing. The contact force is its only horizontal force, so it reads off directly.

$$N_{\text{contact}} = m_2 a = \frac{F m_2}{m_1 + m_2}$$

→ equal masses: each carries half. A very light front block carries almost nothing.

3

Why the same trick keeps working

Ropes, wedges and stacked blocks all obey it: sum over the system to get a, then cut the system where the unknown force lives. The cut replaces the interaction with the unknown.

→ with three bodies you may need two cuts, but the recipe is unchanged

4

A worked contact force

F = 30 N pushes m_1 = 2 kg into m_2 = 4 kg on a frictionless floor. Find a and the contact force.

$$a = \frac{30}{6} = 5 \text{ m/s}^2$$

$$N = m_2 a = 20 \text{ N}, \qquad \text{check: } 30 - N = 2(5)$$

→ the check is one line, and it always catches a wrong mass in the denominator

GATE

the recipe

1 system for a 2 isolate for N 3 check with the other body

KEY RESULT

contact force

$$N = \frac{F m_2}{m_1 + m_2}$$

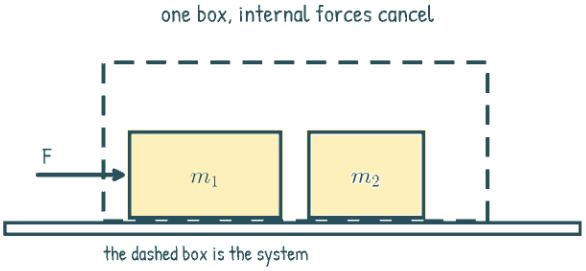
N is not F unless the front block is the whole mass

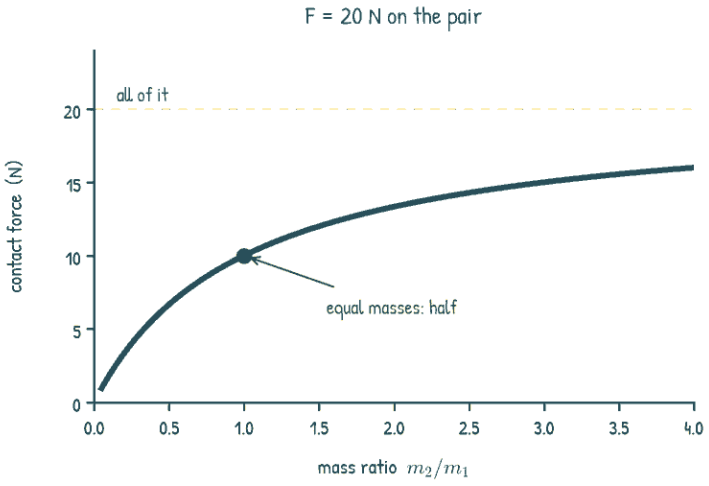
TRAP

adding contact forces to the system

Inside the dashed box the contact forces cancel. If you kept them, you would double count and halve the acceleration.

Use the system for the acceleration, then isolate one body for the force between them





KEEP / use the system for the acceleration, then isolate one body for the internal force

Atwood on a ramp: where does the motion change hands?

? One mass hangs, one sits on a rough 30° ramp. Over what range of mass ratio does nothing move at all?

1 ## The clean Atwood machine

Same tension both sides, same magnitude of acceleration. Adding the two equations eliminates T and gives the standard pair of results.

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2}, \qquad T = \frac{2m_1 m_2 g}{m_1 + m_2}$$

2 ## Put one of them on a ramp

Only the component along the slope drives the ramp mass, so $\sin\theta$ replaces 1 for that side. Add friction and it enters with $\pm$, depending on which way the block is about to slide.

$$a = \frac{(m_2 - m_1\sin\theta)g}{m_1 + m_2} \pm \frac{\mu m_1 g\cos\theta}{m_1 + m_2}$$

3 ## The static band

At rest, friction can supply up to $\mu m_1 g\cos\theta$ in either direction. That gives two thresholds, one for sliding up and one for sliding down, and between them the system is frozen.

$$\sin\theta - \mu\cos\theta \le \frac{m_2}{m_1} \le \sin\theta + \mu\cos\theta$$

→ $\theta = 30°$, $\mu = 0.25$: the band is 0.28 to 0.72

4 ## A worked Atwood

$m_1 = 3$ kg on a smooth 30° ramp, $m_2 = 2$ kg hanging. Find a and T.

$$a = \frac{(2 - 3\sin 30°)9.8}{5} = \frac{0.5(9.8)}{5} = 0.98 \text{ m/s}^2$$

$$T = m_2(g - a) = 2(8.82) = 17.6 \text{ N}$$

→ T is 17.6 N while the hanging weight is 19.6 N: they differ by $m_2 a$

KEY RESULT

the three formulas

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2}$$

$$T = \frac{2m_1 m_2 g}{m_1 + m_2}$$

$$\text{band} : \sin\theta \mp \mu\cos\theta$$

NUMBERS

the worked band

$\theta = 30°$, $\mu_s = 0.25$

0.5 − 0.217 = 0.28 to 0.5 + 0.217 = 0.72

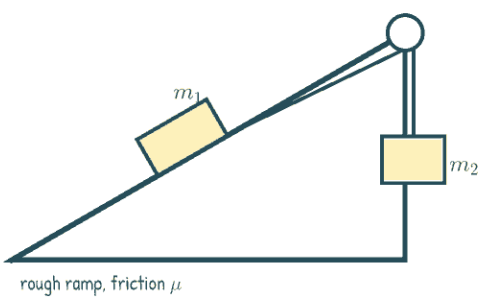
TRAP

T = m g

T equals the hanging weight only when the system is in equilibrium. Otherwise T is found from either body after a is known.

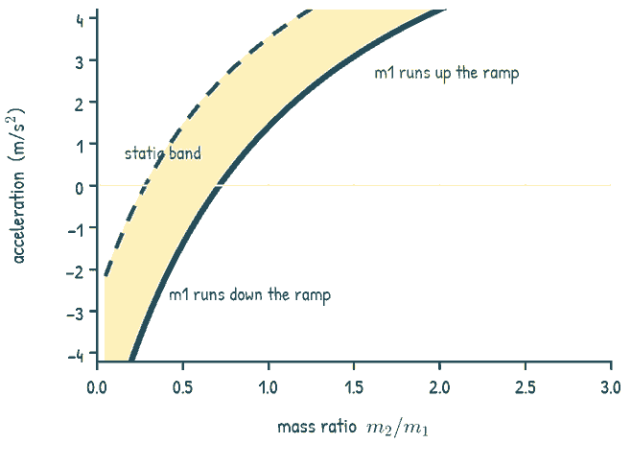
Add friction to the Atwood-on-a-ramp and the acceleration crosses zero at two different mass ratios

one mass on the ramp, one hanging



rough ramp, friction μ

three regimes, one formula



KEEP / friction turns one crossing point into a band of mass ratios

Why does the free end move twice as far?

? A rope runs over a fixed pulley, round a movable one, and is anchored. What links the accelerations, and what links the tensions?

1 ## Write the length, then differentiate

Let x_1 be the segment that ends at the free mass and x_2 the length of each of the two strands holding the movable pulley. The total length is fixed.

$$x_1 + 2x_2 = \text{const} \;\Rightarrow\; a_1 = -2a_2$$

→ the free end must move twice as far, and accelerate twice as hard

2 ## One string, one tension

A massless frictionless pulley cannot support a tension difference, so the rope tension is the same everywhere. The movable pulley therefore feels 2T upward against the weight hanging on it.

$$m_1 g - T = m_1 a_1, \qquad m_2 g - 2T = m_2 a_2$$

3 ## Solve with the constraint

Substitute $a_2 = -a_1/2$ and eliminate T. The free mass always wins the sign argument, because its weight is compared with twice the other weight.

$$a_1 = \frac{2(2m_1 - m_2)g}{4m_1 + m_2}$$

→ equal masses: free end falls at 2g/5, movable pulley rises at g/5

4 ## A worked movable pulley

m_1 = 4 kg at the free end, m_2 = 3 kg on the movable pulley. Find both accelerations and T.

$$a_1 = \frac{2(8-3)9.8}{16+3} = \frac{98}{19} = 5.16\ \text{m/s}^2$$

$$a_2 = -2.58\ \text{m/s}^2, \qquad T = m_1(g - a_1) = 18.6\ \text{N}$$

→ the pulley rises at half the rate the free end falls, and 2T balances m_2 only at rest

KEY RESULT

constraint pair

$$x_1 + 2x_2 = \text{const}$$

$$a_1 = -2a_2, \qquad T \text{ uniform}$$

NUMBERS

equal masses

$m_1 = m_2 = m$

$a_1 = 2g/5$ down, $a_2 = g/5$ up

$T = 3mg/5$

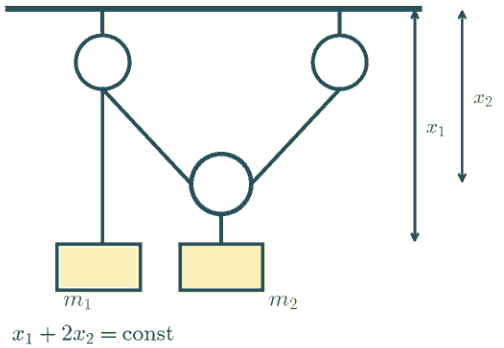
TRAP

sign of the constraint

Decide once which direction is positive for each body. The printed solutions of this chapter are inconsistent about it; the magnitudes 2 : 1 are not.

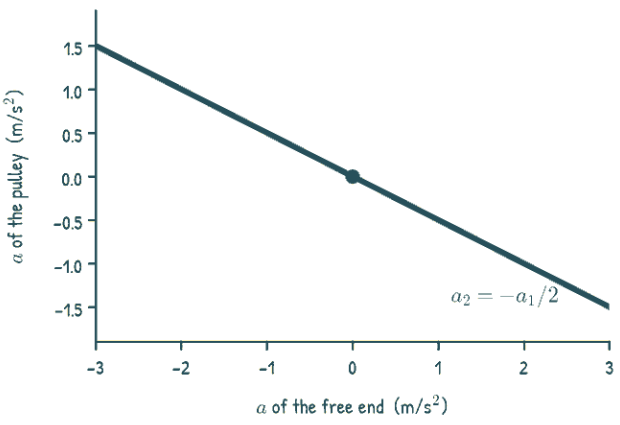
The string length is x1 + 2x2, so the free end accelerates twice as hard as the pulley

one string, three segments



$x_1 + 2x_2 = \text{const}$

differentiate twice



KEEP / the length constraint gives the accelerations; one string gives the tension

Should you ever add a force that is not there?

? In a braking bus a hanging pendulum swings forward. Add a pseudo force, or stay in the ground frame?

1 ### The recipe for an accelerating frame

If the frame accelerates at a_0, every body in it behaves as if an extra force $-ma_0$ acted. Add that one term and Newton's laws work again, with no other changes.

$$\vec{F}_{\text{real}} + (-m\vec{a}_0) = m\vec{a}_{\text{frame}}$$

2 ### The pendulum in the car

In the car frame the bob is at rest with three forces: weight down, pseudo force backward, tension along the string. Resolving gives the hang angle and the tension immediately.

$$\tan\theta = \frac{a_0}{g}, \qquad T = m\sqrt{g^2 + a_0^2}$$

3 ### The ground frame must agree

In the ground frame the bob accelerates forward at a_0. The only real forces are T and mg, and their resultant is horizontal. The same θ falls out — a good check that you have not mixed frames.

$$T\sin\theta = ma_0, \qquad T\cos\theta = mg$$

→ if the two frames disagree, one of them has a force that does not exist

4 ### A worked pendulum in a car

A bob hangs from the roof of a car accelerating at 4.9 m/s². Find the hang angle and the tension per unit mass.

$$\tan\theta = \frac{4.9}{9.8} = 0.5 \;\Rightarrow\; \theta = 26.6^\circ$$

$$T/m = \sqrt{g^2 + a_0^2} = 10.96 \text{ N/kg}$$

→ the same numbers fall out of the ground frame, where T sinθ = ma_0 does the work

KEY RESULT

effective gravity

$$\vec{g}_{\text{eff}} = \vec{g} - \vec{a}_0$$

$$\tan\theta = a_0/g$$

TRAP

mixing frames

The classic disaster: a pseudo force in one equation and ground-frame reasoning in the next. Choose a frame, write it at the top of the page, and stay there.

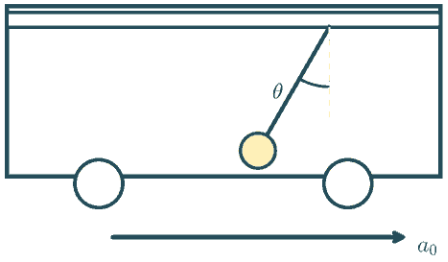
CHECK

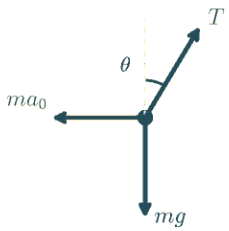
the two-frame test

Solve once each way. Any disagreement is an algebra error, never new physics.

In an accelerating frame, add -ma and then use ordinary equilibrium

accelerating car: the bob hangs askew





$$\tan\theta = a_0/g$$

$$g_{\text{eff}} = \sqrt{g^2 + a_0^2}$$

in the car frame

add the pseudo force

the string lines up with g_eff

KEEP / a pseudo force is frame bookkeeping, not an interaction

Is centripetal force a new force?

? A ball on a string, a car on a bank, a car over a crest. Which real force points inward in each case?

1 ### It is a name, not an interaction

Circular motion at constant speed needs a resultant of mv²/R pointing at the centre. That resultant is produced by whatever real forces happen to point inward. Writing it as an extra force double counts the physics.

$$\sum F_{\text{inward}} = \frac{mv^2}{R} = m\omega^2 R$$

2 ### Three suppliers

A ball on a string: the tension. A car on a bank: the horizontal component of the normal force. A car crossing a hill crest: the difference between weight and normal force, which is why the car feels light.

$$\text{crest}: mg - N = \frac{mv^2}{R}$$

→ at the speed √(gR) the normal force reaches zero and the car leaves the road

3 ### When the supplier quits

A string cannot push, so if the required inward force exceeds the tension available the string goes slack. A tyre cannot pull downward, so on a crest it simply loses contact. The condition is always "the required inward force must be available".

4 ### A worked crest

A car crosses a hill of radius 40 m at 15 m/s. What fraction of its weight does the road still support?

$$N = m\left(g - \frac{v^2}{R}\right) = m(9.8 - 5.63) = 4.17m$$

$$\frac{N}{mg} = 0.43, \qquad v_{\text{lift}} = \sqrt{gR} = 19.8 \text{ m/s}$$

→ at 19.8 m/s the normal force vanishes and the wheels leave the road

DEFINE

centripetal force

The inward component of the resultant, of magnitude mv²/R. It is supplied by real forces; it is not one of them.

TRAP

the extra arrow

Draw mv²/R on the free-body diagram and every equation becomes wrong by that amount. Put it on the other side of the equals sign.

CHECK

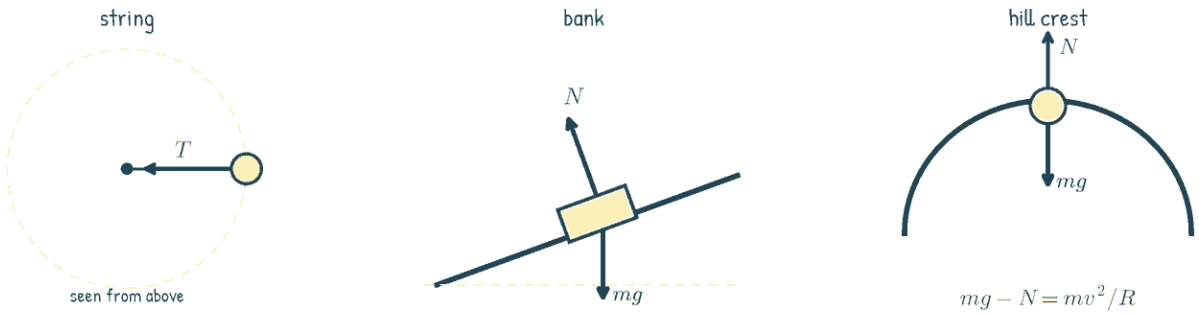
availability test

Compute the inward force required. Compare it with the largest inward force the contact can supply. If required > available, contact is lost.

Centripetal force is the name of the inward resultant, never an extra arrow

$$\sum F_{\text{inward}} = mv^2/R$$

the resultant is named, not added



KEEP / centripetal force is the inward resultant, never an extra arrow

Why are racetrack corners banked?

? With no friction at all, is there still a speed at which a car can take a banked curve?

1 ## Resolve vertically and radially

On a frictionless bank the only forces are weight and the normal, which is tilted. Tilt means it has an inward component, and that component is the centripetal force.

$$N\cos\theta = mg, \qquad N\sin\theta = \frac{mv^2}{R}$$

2 ## Divide: the mass disappears

Dividing the two equations removes both m and N, leaving the design speed. Every vehicle, from a bicycle to a truck, needs the same speed for a frictionless passage.

$$v_0^2 = Rg\tan\theta$$

→ R = 50 m, θ = 22°: v_0 = 14 m/s, about 50 km/h

3 ## Why real roads still need friction

Traffic does not travel at exactly v_0. Below it the car tends to slide down the bank, above it up the bank, and friction has to supply the difference. That is the width of the band on the next sheet.

$$N = \frac{mg}{\cos\theta} > mg$$

4 ## A worked bank

A curve of radius 80 m is banked for 72 km/h with no friction. Find the angle and the normal force on a 1200 kg car.

$$\tan\theta = \frac{v^2}{Rg} = \frac{400}{784} = 0.51 \Rightarrow \theta = 27^\circ$$

$$N = \frac{mg}{\cos\theta} = \frac{11760}{0.891} = 13.2\ \text{kN}$$

→ N exceeds the weight by 12 %, because the bank must also turn the car

KEY RESULT

design speed

$$v_0 = \sqrt{Rg\tan\theta}$$

$$\tan\theta = \frac{v_0^2}{Rg}$$

NUMBERS

typical corner

R = 50 m, θ = 22°, g = 9.8

v_0 = 14.0 m/s = 50.5 km/h

TRAP

N = mg on a bank

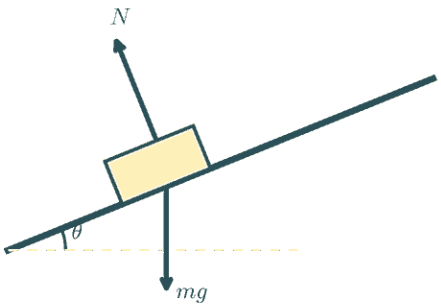
The normal force is larger than mg on a bank: N = mg/cosθ. Using N = mg understates the available friction as well.

With no friction there is exactly one speed that needs no friction at all

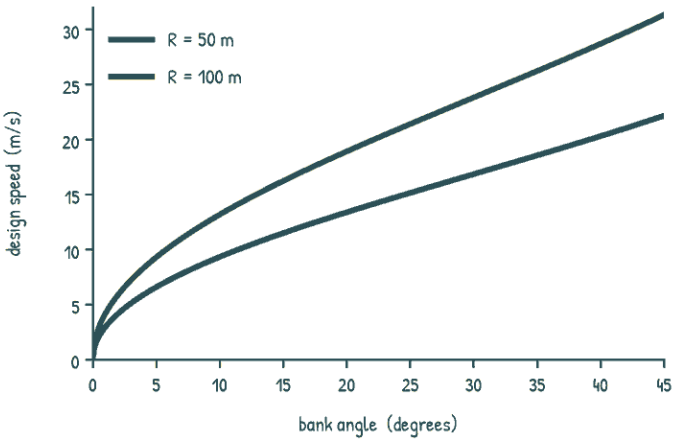
$$N\cos\theta = mg$$

$$N\sin\theta = mv^2/R$$

$$v^2 = Rg\tan\theta$$



steeper bank, faster design speed



KEEP / a bank turns normal force into centripetal force without any friction

How much slower or faster can you take the bank?

? Friction turns one design speed into a range. What are the two limits, and what sets them?

1 ## Two limits, one derivation

At the maximum speed the car is about to slide outward, so friction points down the slope. At the minimum it is about to slide inward, so friction points up the slope. Resolve both cases the same way.

$$v_{\max}^2 = \frac{Rg(\sin\theta + \mu\cos\theta)}{\cos\theta - \mu\sin\theta}$$

$$v_{\min}^2 = \frac{Rg(\sin\theta - \mu\cos\theta)}{\cos\theta + \mu\sin\theta}$$

2 ## The three regimes

Below v_min friction points up the slope; at v_0 it is exactly zero; above v_0 it points down. The same formulas cover all three, and v_min is set to zero once μ exceeds tanθ.

→ if μ > tanθ the car can stand still on the bank with the brakes off

3 ## Reading the formulas safely

The denominator of v_max vanishes at μ = cotθ, where no speed is safe. That is not a curiosity: it is why very steep banks are only used with high-grip tyres and why ice on a banked road is so dangerous.

4 ## A worked band

Same corner: R = 80 m, θ = 27°, μ = 0.35. Find the fastest and slowest safe speeds.

$$v_{\max}^2 = \frac{784(0.454 + 0.312)}{0.891 - 0.159} = 820 \;\Rightarrow\; 28.6\ \text{m/s}$$

$$v_{\min}^2 = \frac{784(0.454 - 0.312)}{0.891 + 0.159} = 106 \;\Rightarrow\; 10.3\ \text{m/s}$$

→ the design speed sits between them at 20 m/s, as it must

KEY RESULT

the band

$$v_{\max}^2 = \frac{Rg(\sin\theta + \mu\cos\theta)}{\cos\theta - \mu\sin\theta}$$

$$v_{\min}^2 = \frac{Rg(\sin\theta - \mu\cos\theta)}{\cos\theta + \mu\sin\theta}$$

TRAP

sign in the denominator

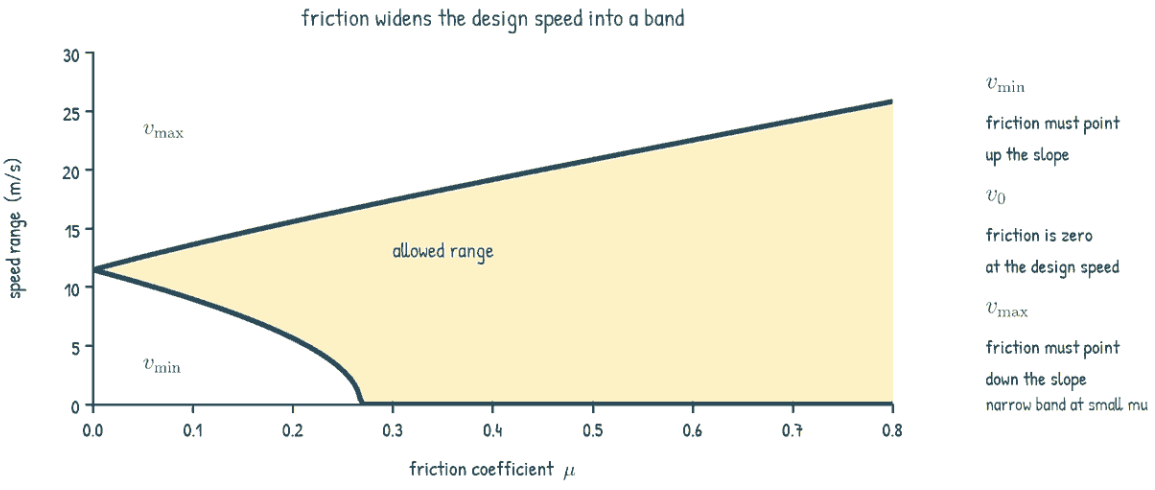
The two denominators differ: cosθ − μ sinθ for the fast limit, cosθ + μ sinθ for the slow one. Swapping them still gives plausible numbers.

CHECK

set mu = 0

Both formulas must collapse to $v_0 = \sqrt{Rg\tan\theta}$. If they do not, the algebra is wrong.

Friction does not set the design speed; it allows a range of speeds around it



KEEP / friction widens a single design speed into a band of allowed speeds

Why does a conical pendulum ignore the mass?

? A bob whirled on a string traces a horizontal circle. What actually sets the period?

1 **Separate vertical from radial**

The string makes a constant angle θ with the vertical. Vertically there is no acceleration; radially the horizontal component of the tension is the centripetal force.

$$T\cos\theta = mg, \qquad T\sin\theta = m\omega^2(L\sin\theta)$$

2 **Divide and read off the period**

The sinθ cancels, and so does the mass. What survives is that the bob behaves like a simple pendulum of length L cosθ — the vertical drop from the support to the plane of the circle.

$$\omega^2 = \frac{g}{L\cos\theta}, \qquad T_p = 2\pi\sqrt{\frac{L\cos\theta}{g}}$$

→ a faster whirl means a larger θ, not a different length

3 **The two limits**

As θ → 0 the circle shrinks and the period tends to the ordinary pendulum 2π√(L/g). As θ → 90° the required ω diverges, which is why you cannot whirl a bob into a horizontal circle.

4 **A worked whirl**

A 0.5 m string makes 40° with the vertical. Find the speed of the bob and the tension in a 0.2 kg bob.

$$T = \frac{mg}{\cos 40^\circ} = \frac{1.96}{0.766} = 2.56\ \text{N}$$

$$v = \sqrt{gL\sin\theta\tan\theta} = \sqrt{9.8(0.5)(0.643)(0.839)} = 1.63\ \text{m/s}$$

→ the period 2π√(L cosθ/g) = 1.24 s is independent of the mass

KEY RESULT

the result

$$T_p = 2\pi\sqrt{\frac{L\cos\theta}{g}}$$

$$\cos\theta = \frac{g}{\omega^2 L}$$

TRAP

using L sin theta

The radius is L sinθ, but the period uses the vertical drop L cosθ. Mixing them gives a period that grows with θ instead of shrinking.

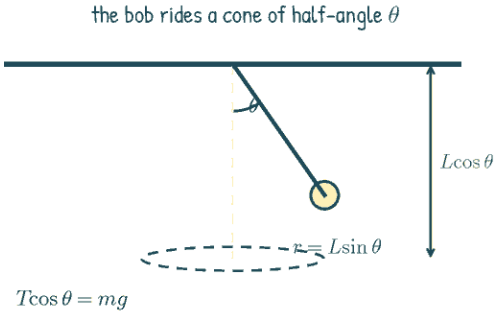
NUMBERS

a check

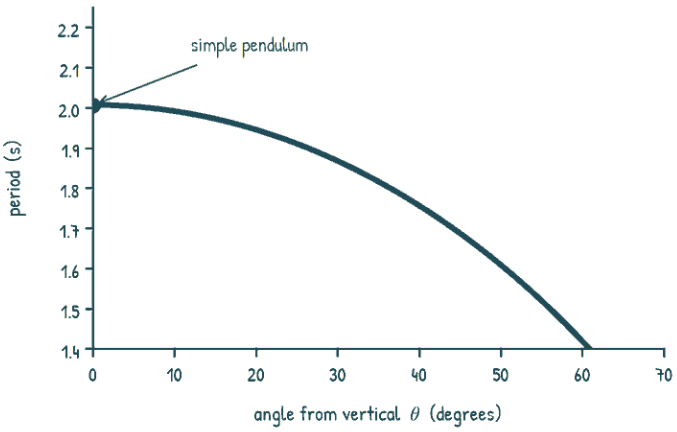
L = 1 m, θ = 60°: T = 2π√(0.5/9.8) = 1.42 s

the simple pendulum would give 2.01 s

The vertical component carries the weight, so the period depends on L cos(theta), never on mass



it shortens as the cone opens



KEEP / the vertical component carries the weight, so only L cos theta enters the period

What is the slowest a ball can go round a vertical loop?

? A ball on a string is whirled in a vertical circle. Why must it be fast at the bottom, and what happens at the top?

1 ## The string can pull but not push

At the top both the tension and the weight point downward, toward the centre. At the limiting speed the tension just reaches zero and the weight alone supplies mv²/R.

$$T + mg = \frac{mv^2}{R} \;\Rightarrow\; v_{\text{top, min}} = \sqrt{gR}$$

2 ## Energy carries that back to the bottom

Going from the bottom to the top costs 2mgR. Put in the minimum top speed and the required launch speed falls out; then the tension at any angle follows from the radial equation.

$$v_{\text{bot, min}} = \sqrt{5gR}$$

$$T(\theta) = 3mg(1 + \cos\theta)$$

→ θ measured from the bottom: 6mg at the bottom, 0 at the top

3 ## Losing the string

Launch slower than √(5gR) and the string slackens before the top. The ball then moves as a projectile inside the circle, and it re-tensions with a jerk if the geometry brings it back to the string.

→ the slack angle is where T = 0 and the radial equation still holds

4 ## A worked loop

A 0.4 kg ball on a 0.8 m string is whirled at exactly the limiting speed. Find the launch speed and the tension at the bottom.

$$v_{\text{bot}} = \sqrt{5gR} = \sqrt{5(9.8)(0.8)} = 6.26 \text{ m/s}$$

$$T_{\text{bot}} = 6mg = 6(3.92) = 23.5 \text{ N}$$

→ the same ball at the top has T = 0 and v = 2.8 m/s, not zero

KEY RESULT

the limiting launch

$$v_{\text{top}} = \sqrt{gR}$$

$$v_{\text{bot}} = \sqrt{5gR}$$

$$T(\theta) = 3mg(1 + \cos\theta)$$

NUMBERS

the six and the zero

T_bottom = 6mg, T_top = 0

at the horizontal position T = 3mg

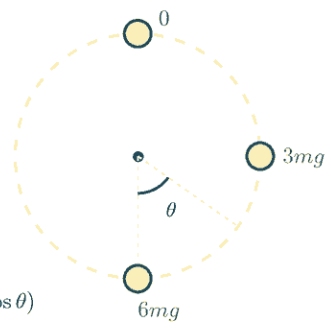
TRAP

v_top = 0

Zero speed at the top is the rod condition, not the string condition. On a string the ball must still be moving fast enough for mg to supply mv²/R.

At the limiting launch the string is just taut at the top: six mg at the bottom, zero at the top

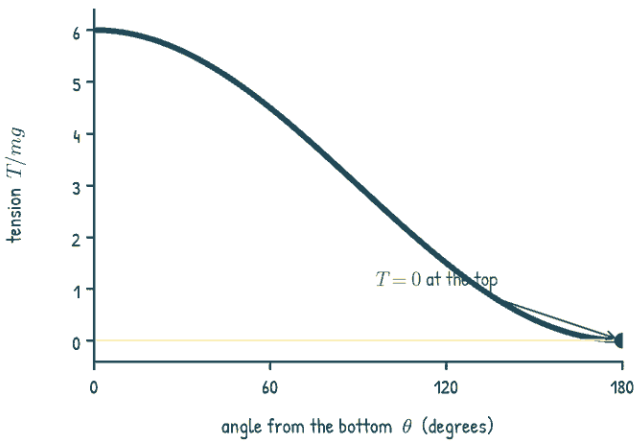
a string at the limiting speed



$T(\theta) = 3mg(1 + \cos\theta)$

$v_{\text{bot}} = \sqrt{5gR}$

tension falls all the way up



KEEP / the string can only pull, so at the top it must still be taut

Why can a bead on a rod stop at the top?

? A bead threaded on a light rod is given just enough speed to reach the highest point. What is different from the string?

1 ### A rod is a two-sided constraint

The rod can exert a force toward the centre or away from it. That means v can fall to zero at the top without the constraint failing, and the threshold √(gR) simply disappears.

$$N + mg = \frac{mv^2}{R} \quad \text{with} N \text{ of either sign}$$

2 ### Energy fixes the speed everywhere

Give the bead just enough to reach the top, v_top = 0, so v_bot = √(4gR) = 2√(gR). Then the radial equation gives N at any angle, with cosθ measured from the bottom.

$$N(\theta) = mg(2 + 3\cos\theta)$$

→ 5mg at the bottom, −mg at the top: the rod is pulling, not pushing

3 ### The sign flip

Set N = 0 and cosθ = −2/3, which is 131.8°. Below that the rod pushes outward; above it the rod pulls inward. The bead is never in danger of leaving the circle.

4 ### A worked rod loop

A 0.1 kg bead on a 0.6 m rod is given just enough speed to reach the top. Find the launch speed and N at the horizontal position.

$$v_{\text{bot}} = 2\sqrt{gR} = 2\sqrt{5.88} = 4.85 \text{ m/s}$$

$$N(90^\circ) = 2mg = 1.96 \text{ N inward}$$

→ only half the string's launch speed, because the rod does not need tension at the top

KEY RESULT

rod results

$$v_{\text{bot, min}} = 2\sqrt{gR}$$

$$N(\theta) = mg(2 + 3\cos\theta)$$

$$N = 0 \text{ at } \cos\theta = -2/3$$

TRAP

string and rod mixed

The exam favourite: check whether the constraint can push. If it can, the top speed may be zero; if it cannot, the minimum is √(gR).

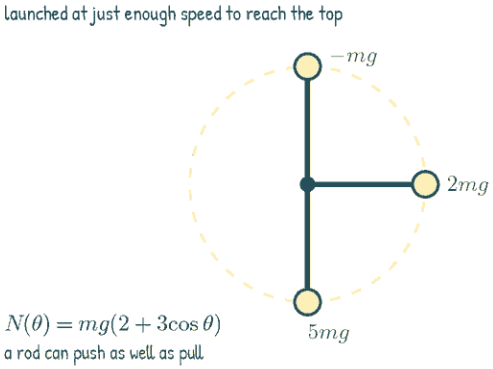
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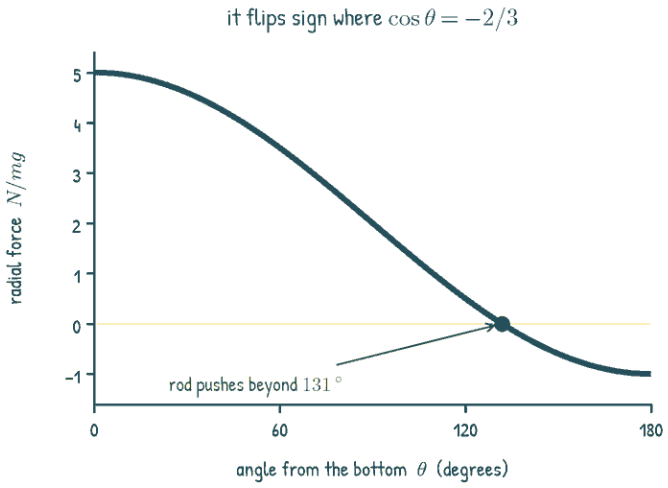
sign check

bottom: N = 5mg (outward, rod pushes)

top: N = −mg (inward, rod pulls)

A rod needs no minimum speed at the top, and past 131 degrees it stops pulling and starts pushing





KEEP / a rod pushes as well as pulls, so it needs no minimum speed at the top

Why is energy faster than forces in a vertical circle?

? The tangential equation has a sin theta in it and the radial equation has a v squared. How do you avoid integrating at all?

1 Two equations, two different jobs

The radial equation relates T and v at one instant. It says nothing about how v changes with angle. For that, use energy: weight is conservative, so the speed at any height follows from the launch speed.

$$v^2(\theta) = v_0^2 - 2gR(1+\cos\theta)$$

→ θ is measured from the bottom, so the rise is R(1 + cosθ)

2 Then one radial equation finishes it

Substitute the speed into the radial relation. The tension now comes out as a function of angle with no integration anywhere.

$$T(\theta) = m\frac{v^2(\theta)}{R} - mg\cos\theta$$

$$T(\theta) = \frac{mv_0^2}{R} - mg(2+3\cos\theta)$$

→ this single formula contains the string results 6mg and 0 at the two ends

3 Why the tangential route is a trap

You could write the tangential equation a_t = −g sinθ and integrate it with v dv = a_t R dθ. It gives the same answer, in three times the lines and with one more place to lose a sign.

→ the energy route also tells you at a glance which launches are too slow

4 A worked energy check

A 0.5 kg ball on a 1 m string is launched at 8 m/s. Use energy to find the speed at the top and the tension there.

$$v_{\text{top}}^2 = 64 - 4(9.8)(1) = 24.8 \;\Rightarrow\; v = 4.98\ \text{m/s}$$

$$T = \frac{0.5(24.8)}{1} - 0.5(9.8) = 7.5\ \text{N}$$

→ the tangential integral would give the same 24.8, in three more lines

KEY RESULT

the pair

$$v^2 = v_0^2 - 2gR(1+\cos\theta)$$

$$T = \frac{mv^2}{R} - mg\cos\theta$$

NUMBERS

a check with v0 = 8 m/s

R = 1 m, m = 0.5 kg

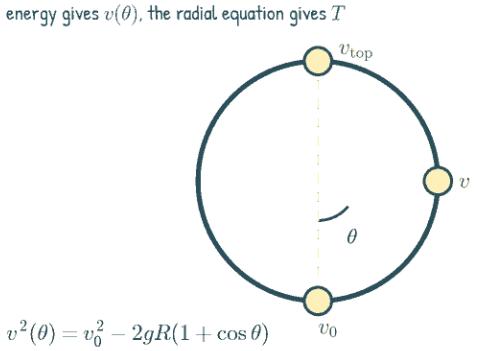
bottom T = 37 N, side 22 N, top 7 N

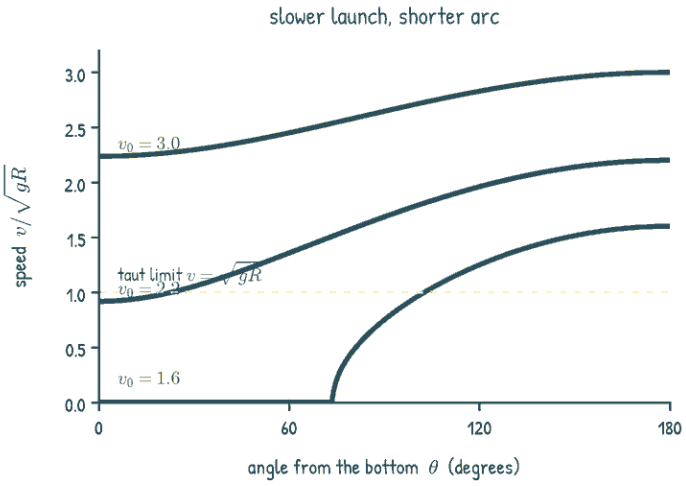
TRAP

the sign of mg cos theta

At the top the weight points inward, so the radial equation is T + mg = mv²/R. Written with cosθ from the bottom, that is exactly the minus sign above.

Energy gives the speed at every angle in one line, and the radial equation does the rest





KEEP / energy gives the speed at every angle; the radial equation gives the tension

What changes when the table is rough?

? A block on a rough table is tied over a pulley to a hanging mass. Everything is the same as the smooth case except one term. Which one?

1 **Both bodies, one string**

The tension is the same throughout and the magnitudes of the accelerations match. For the table block the tension pulls forward and friction holds back; for the hanging mass the weight pulls down and the tension up.

$$T - \mu_k m g = m a, \qquad M g - T = M a$$

2 **Eliminate T**

Adding the two equations removes the tension and gives the acceleration directly. Friction enters only once, as a subtraction from the drive.

$$a = \frac{(M - \mu_k m) g}{M + m}$$

→ set μ_k = 0 and the standard smooth result returns at once

3 **The two thresholds**

At rest the system stays put while M g ≤ μ_s m g. That gives a maximum hanging mass. Once moving, the acceleration is positive only if M > μ_k m, which is a slightly lower threshold.

$$M_{\max} = \mu_s m \quad \text{(static)}, \qquad M > \mu_k m \quad \text{(moves)}$$

4 **A worked rough pulley**

m = 4 kg on a table with μ_k = 0.3, M = 3 kg hanging. Find a and T.

$$a = \frac{(3 - 1.2) 9.8}{7} = 2.52 \text{ m/s}^2$$

$$T = M(g - a) = 3(7.28) = 21.8 \text{ N}$$

→ T is below Mg = 29.4 N, because the hanging mass is accelerating down

KEY RESULT

the result

$$a = \frac{(M - \mu_k m) g}{M + m}$$

$$T = \frac{M m (1 + \mu_k) g}{M + m}$$

NUMBERS

worked

m = 3 kg, M = 2 kg, μ_k = 0.4

a = (2 − 1.2)(9.8)/5 = 1.57 m/s²

T = 16.5 N, below the hanging weight

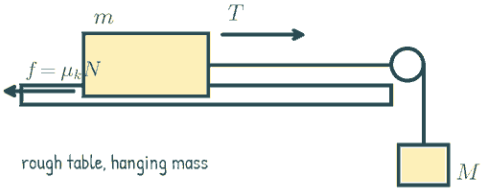
TRAP

which mu

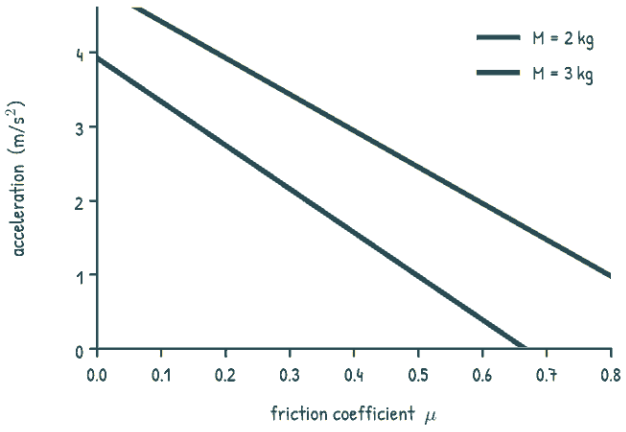
Use μ_s only for the "does it start?" question and μ_k once it moves. Mixing them gives a system that accelerates from rest with the kinetic value.

On the table friction is the brake, on the string it is the drive: separate books

one string, two different jobs



m = 3 kg on the table



KEEP / on the table friction is the brake; on the string it is the drive

Can an accelerating ramp hold a block still?

? A block sits on a frictionless ramp that is itself accelerating horizontally. Is there a ramp acceleration at which the block does not slide at all?

1 Work in the ramp frame

The block feels weight down, normal perpendicular to the face, and the pseudo force backward. Resolve along the face: the two driving terms are the downhill part of gravity and the uphill part of the pseudo force.

$$ma_{\text{rel}} = mg\sin\theta - ma_0\cos\theta$$

2 Set the relative acceleration to zero

One line gives the balanced ramp acceleration. It depends on nothing but the angle — not on m, not on g separately.

$$a_0 = g\tan\theta$$

→ θ = 30°: a_0 = 5.66 m/s² holds the block in place

3 Then the normal force

Resolve perpendicular to the face. Both the weight component and the pseudo component push the block into the ramp, so accelerating the ramp into the block increases N rather than threatening contact.

$$N = mg\cos\theta + ma_0\sin\theta > 0$$

→ this is a stability threshold, not a loss-of-contact threshold

4 A worked accelerating ramp

A 2 kg block rests on a 35° smooth ramp. Find the ramp acceleration that keeps it there and the normal force then.

$$a_0 = g\tan 35^\circ = 9.8(0.700) = 6.86\ \text{m/s}^2$$

$$N = \frac{mg}{\cos 35^\circ} = \frac{19.6}{0.819} = 23.9\ \text{N}$$

→ N is 22 % above the weight; the ramp is being pushed into the block

KEY RESULT

the balance

$$a_0 = g\tan\theta$$

$$N = m(g\cos\theta + a_0\sin\theta)$$

TRAP

confusing with liftoff

Pushing the ramp into the block raises N. Pulling the ramp away lowers it. Only the second case has a liftoff threshold, and it is a different formula.

NUMBERS

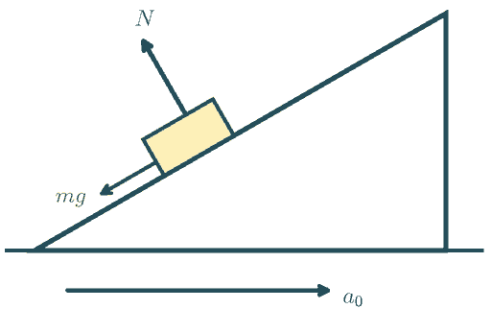
the two angles

θ = 30°: hold at 5.66 m/s²

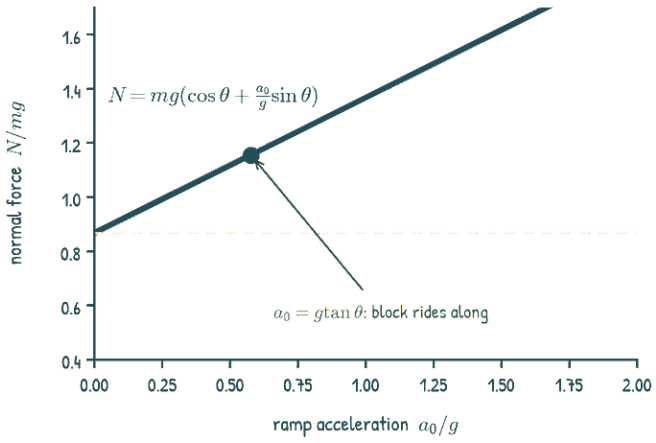
θ = 45°: hold at 9.8 m/s²

Push the ramp into the block and N rises; the block rides at rest when a0 = g tan(theta)

the ramp itself is accelerating



accelerating the ramp loads the block



KEEP / accelerate the ramp at g tan theta and the pseudo force replaces gravity along the face

How do you measure friction without a force meter?

? Friction coefficients are quoted to two digits. Which three independent experiments produce them, and what does each one assume?

1 ### The tilt measures the static coefficient

Raise one end of a plank until the block just starts to slide. At that instant the drive mg sinθ and the ceiling μ_s mg cosθ are equal, so the coefficient is the tangent of the angle.

$$\mu_s = \tan\theta_s$$

→ μ_s = 0.4 means θ_s = 21.8°, easy to read on a protractor

2 ### The skid measures the kinetic one

Lock the wheels or slide a box and measure the stopping distance from a known speed. The deceleration is μ_k g, so the coefficient follows from kinematics alone.

$$\mu_k = \frac{v^2}{2gd}$$

3 ### The drag measures it a third way

Pull at constant speed and the applied force equals the kinetic friction, so the coefficient is the ratio of two forces. This is the only one of the three that needs no kinematics, and the only one that reads the coefficient at speed.

$$\mu_k = \frac{F}{mg}$$

→ do it at several speeds: real friction is not perfectly constant

4 ### A worked measurement

A box slides 18 m before stopping from 12 m/s. Find μ_k; then find the tilt angle that would just hold it at rest if μ_s is 15 % larger.

$$\mu_k = \frac{144}{2(9.8)(18)} = 0.41$$

$$\mu_s = 0.47 \;\Rightarrow\; \theta_s = \tan^{-1} 0.47 = 25^\circ$$

→ two digits is all the data supports, and all μ is ever good for

KEY RESULT

three formulas

$$\mu_s = \tan\theta_s$$

$$\mu_k = v^2/(2gd)$$

$$\mu_k = F/(mg)$$

NUMBERS

the same surface

tilt: θ_s = 22° gives 0.40

skid from 10 m/s in 13.0 m: 0.39

drag at steady speed: 0.39

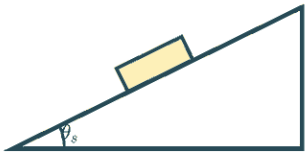
TRAP

the tilt measures mu_s only

A sliding block on the same plank is no longer at the static limit; the tangent of the angle is then just a lower bound on μ_s.

Three independent measurements of one number: tilt, brake, drag

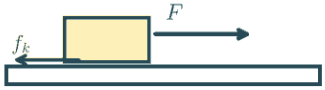
$\mu_s = \tan\theta_s$ $\mu_k = v^2/(2gd)$ $F = \mu_k mg$ at steady v



tilt until it slides



brake and measure d



drag at constant speed

KEEP / three routes to one number: tilt, brake, and drag at constant speed

Why does a chain accelerate faster as it falls?

? A chain hangs over a table edge with length l already off. The driving weight grows as more falls, but the mass being accelerated does not. What motion results?

1. ### Write the force with x in it

The hanging part has weight λxg and the whole chain has mass λL. Friction is ignored here. The acceleration therefore depends on position, which means the motion is not uniform.

$$a = \frac{\lambda x g}{\lambda L} = \frac{g x}{L}$$

2. ### Integrate instead of using a constant-a formula

Use v dv = a dx. The result is a difference of squares, not a linear ramp in time, and it shows the speed saturating only when the last link leaves.

$$v\,dv = \frac{g}{L} x\,dx \;\Rightarrow\; v^2 = \frac{g}{L}(x^2 - l^2)$$

$$v_{x=L} = \sqrt{\frac{g}{L}(L^2 - l^2)}$$

→ set l = 0 and the speed is √(gL) — the classic result

3. ### The same idea with friction

Add μ times the normal force on the part still on the table and the drive becomes x − μ(L − x). The acceleration can start negative: the chain then waits until enough hangs over.

4. ### A worked chain

A 1.5 m chain of mass 0.6 kg has 0.4 m hanging. Find the initial acceleration and the speed when it leaves the table.

$$a = \frac{gx}{L} = \frac{9.8(0.4)}{1.5} = 2.61\ \text{m/s}^2$$

$$v = \sqrt{\frac{9.8}{1.5}(2.25 - 0.16)} = 3.70\ \text{m/s}$$

→ the final speed is close to √(gL) = 3.83 because only 0.4 m was left hanging

KEY RESULT

the motion

$$a = gx/L$$

$$v^2 = \frac{g}{L}(x^2 - l^2)$$

NUMBERS

a check

L = 2 m, l = 0.5 m

v at the end = √(9.8(3.75)/2) = 4.3 m/s

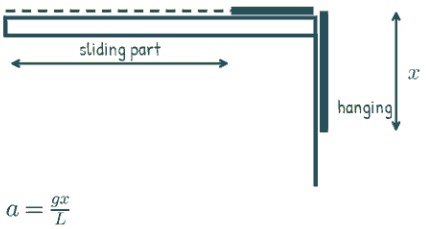
TRAP

constant acceleration

Using a = gx/L as a constant with s = ½at² gives the wrong speed and the wrong time. The acceleration is a function of position.

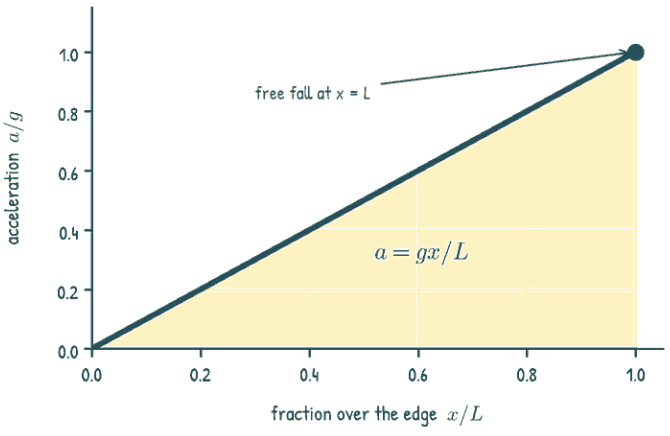
The hanging weight grows while the sliding mass stays the same, so the acceleration grows with x

as x grows, the drive grows



$$a = \frac{gx}{L}$$

a linear ramp, not a constant



KEEP / the hanging weight grows while the sliding mass stays the same, so a grows with x

How hard must you push a wedge to carry its block?

? A block rests on a smooth wedge. Accelerate the wedge horizontally and the block can stay put. What is the required acceleration, and what keeps it there?

1 The two ways to hold the block

On a fixed smooth wedge the block slides down, because nothing balances mg sinθ. Accelerating the wedge adds a pseudo force whose uphill component can balance it exactly.

$$ma_0\cos\theta = mg\sin\theta \;\Rightarrow\; a_0 = g\tan\theta$$

2 What the rider feels

The normal force rises, because the pseudo force also presses the block into the face. That is not a coincidence: the same push that replaces gravity along the slope also loads the contact.

$$N = \frac{mg}{\cos\theta}$$

→ a 1 kg block on a 30° wedge at the balance point presses with 11.3 N, not 8.5 N

3 The applied force on the whole system

Treat wedge plus block as one body. The external push must accelerate the total mass at a_0, and the internal normal does not appear at all.

$$F = (M+m)g\tan\theta$$

4 A worked carry

A 2 kg block on a 40° wedge is to ride along. Find the required acceleration and the push on a 5 kg wedge.

$$a_0 = 9.8\tan 40^\circ = 8.22\ \text{m/s}^2$$

$$F = (5+2)(8.22) = 57.6\ \text{N}, \qquad N = \frac{19.6}{\cos 40^\circ} = 25.6\ \text{N}$$

→ the push scales with the total mass; the balance angle does not

KEY RESULT

the balance

$$a_0 = g\tan\theta$$

$$N = mg/\cos\theta$$

$$F = (M+m)g\tan\theta$$

NUMBERS

worked

M = 4 kg, m = 1 kg, θ = 30°

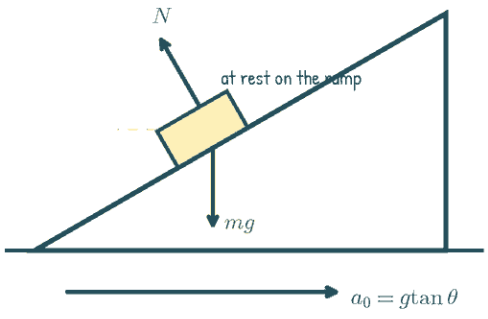
a_0 = 5.66 m/s², F = 28.3 N

TRAP

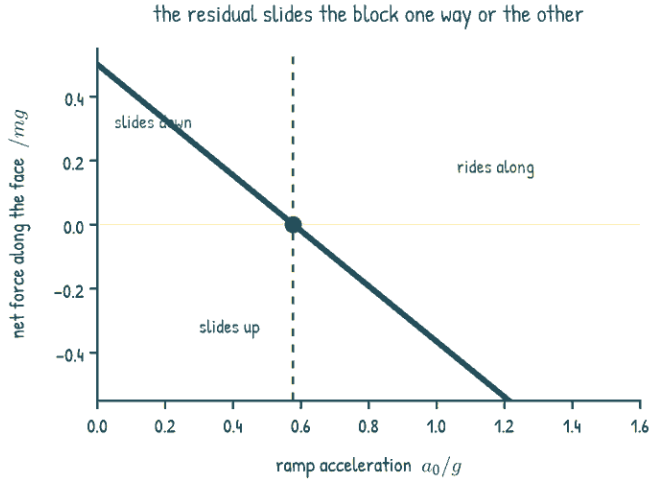
pushing the wedge the other way

Reverse the acceleration and the pseudo force now adds to the downhill pull: no speed can hold the block. Direction is part of the answer.

At a0 = g tan(theta) the pseudo force exactly replaces the downhill pull of gravity



the residual slides the block one way or the other



KEEP / at a0 = g tan theta the pseudo force exactly replaces the downhill pull